

UMAR ODULAJA

Senior Data Scientist

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PROFESSIONAL SUMMARY

Senior Data Scientist with 8 years of experience designing, deploying, and scaling end-to-end machine learning, generative AI, and cloud-native data solutions across financial services, defense, and consumer technology. Track record of architecting LLM-powered products, predictive risk models, and production ETL pipelines that translate ambiguous business problems into measurable, multi-million-dollar outcomes. Expert in Python, SQL, Alteryx, AWS, Snowflake, AI/ML and GenAI, with a proven ability to communicate complex insights to executive stakeholders and drive cross-functional decision-making.

PROFESSIONAL EXPERIENCE

JPMorgan Chase — *Senior Data Scientist* | Sept 2024 – Present

- Built and deployed a production LLM-powered agent that enables business users to query Chase Auto complaints data in natural language and generate data-driven insights without relying on manual SQL queries. Designed the backend architecture using Markdown-based business context, Bitbucket retrieval for historical SQL/code, Claude Sonnet 4.6 for reasoning, and secure Snowflake integration for dynamic SQL generation and data retrieval, accelerating analyst time-to-insight by 70% and reducing ad-hoc reporting volume by 40%.
- Engineered an internal multi-agent AI system to automate the team's Jira request workflow end-to-end, reducing manual effort by ~2 hours per request per team member. Built specialized sub-agents to update Jira fields, generate change-request documentation, push code to Bitbucket, and create pull requests, streamlining intake, documentation, and delivery.
- Developed NLP and LLM-driven analytics capabilities to classify and extract insights from unstructured auto-complaint data, improving taxonomy labeling accuracy by 60% and accelerating root-cause identification. Surfaced ~\$4M in annual loss drivers from customer text data, shaping a digital payments solution that improved customer experience, increased digital adoption, and reduced financial losses.
- Built machine learning models to identify high-risk account behavior, anomaly patterns, and potential fraud signals across Chase's auto loan and lease portfolios, improving proactive detection and reducing financial loss exposure across a multi-billion-dollar portfolio.
- Designed and delivered a scalable customer analytics product that integrated multiple enterprise data sources into executive-ready Tableau dashboards, giving leadership a centralized view of customer trends, operational issues, and business performance to support faster, data-driven decision-making.
- Provided technical mentorship to 3 junior data scientists while implementing peer code review practices, CI/CD checks, and deployment standards that improved team engineering quality and reduced model deployment cycle time by 35%.

Lockheed Martin — *Senior Data Analyst* | May 2022 – Sept 2024

- Built and productionized a Python-based ETL pipeline to automate ingestion, transformation, and delivery of F-35 sustainment data into Tableau, reducing batch-processing time by 90% while improving operational visibility into material shortages, cost drivers, and sustainment planning.
- Developed and maintained interactive Tableau dashboards for 150+ monthly users across multiple F-35 production teams, providing leadership with tiered operational metrics to improve oversight, monitor performance, and make more informed decisions while reducing data response time by 50%.

- Designed and implemented a secure AWS-based relational data platform that centralized analytics data for 50+ daily users, leveraging IAM access controls, encryption at rest, and CloudWatch monitoring to improve data governance, reliability, and secure self-service access.
- Developed a predictive forecasting model for quarterly and annual spend planning, enabling leadership to forecast expenditures, identify budget risks, and strategically allocate funds across a \$500M F-35 sustainment budget.

Lockheed Martin — *Manufacturing Engineer Asc/Aeronautical Engineer* | April 2018 – May 2022

- Resolved 50+ aircraft engineering issues across multiple F-16 programs by analyzing engineering drawings and 3D CAD models to identify root causes, implement design changes, and release revised technical documentation for aircraft modification.
- Partnered with engineering, manufacturing, and quality stakeholders to execute design changes, translating complex technical findings into clear recommendations for both technical and non-technical audiences.
- Led cross-functional process improvement projects that used data-driven root-cause analysis and corrective action plans to improve manufacturing workflows, present recommendations to leadership, and secure producibility funding.

TECHNICAL SKILLS

- **Languages:** Python, SQL
- **ML & Deep Learning:** Scikit-Learn, MLflow, XGBoost, Predictive Modeling, Feature Engineering
- **Gen AI & LLMs:** Claude, OpenAI/ChatGPT, RAG, Agentic Workflows, Prompt Engineering, Embeddings, Vector Databases, Fine-Tuning, MCP, NLP Classification, Text Analytics, Model Evaluation
- **Data & Cloud:** Snowflake, Alteryx, PostgreSQL, Oracle, AWS (SageMaker, Bedrock, Lambda, Glue, S3)
- **MLOps & DevOps:** Docker, Kubernetes, Git/GitHub Actions, CI/CD, Model Deployment & Monitoring
- **Statistics:** A/B Testing, Statistical Modeling, Time Series Forecasting, Anomaly Detection
- **Visualization & Storytelling:** Tableau, Power BI, Streamlit, Plotly
- **Tools & Methodologies:** Jira, Confluence, Monday, Agile/Scrum, Stakeholder Management

AI PROJECTS

- Built an AI fitness app that uses OpenAI GPT 5 Model to generate personalized workout plans based on user goals, exercise history, and performance trends. Designed features for exercise tracking, progression monitoring, and goal-based recommendations to deliver adaptive, data-driven training plans.
- Created an AI-powered scheduling app that converts daily to-do lists into personalized, time-blocked schedules, with interactive timeline editing, completion tracking, and an online-learning engine that improves future schedules based on historical user behavior.
- Built an automated playlist curation system leveraging the Spotify API and Claude to discover, score, and add weekly new releases that best match a playlist's evolving sonic identity, with a three-stage discovery pipeline and GitHub Actions scheduling for fully autonomous updates.
- Developed an XGBoost machine-learning pipeline to predict NBA player stats, applying advanced feature engineering and model evaluation techniques to iteratively improve forecast accuracy.

CERTIFICATIONS

- AWS Certified AI Practitioner
- DataCamp Certified Data Scientist
- IBM Data Science Professional Certificate

EDUCATION

- **Southern Methodist University — *M.S. Systems Engineering* | 2019 – 2022**
- **Baylor University — *B.S. Mechanical Engineering, Minor in Mathematics* | 2013 – 2017**